

Ind Fairness in IM

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Proposed Method

The Myopic algorithm selects the node with the lowest probability of receiving information. Although this directly targets the least-accessible individual, it does not consider whether the selected node has sufficient spreading capability to improve information access for other disadvantaged nodes. In particular, a node with a very low probability of receiving information may be located at the network periphery and may have limited ability to propagate information further. To address this limitation, we propose a simple propagation-aware framework that considers two complementary characteristics of each candidate node:

1. **Accessibility disadvantage:** How unlikely is the node to receive the information?
2. **Spreading power:** If selected as a seed, how effectively can the node propagate information to other disadvantaged nodes?

The two quantities are combined to obtain a fairness-aware seed selection score.

1.1 Step 1: Estimating Information Accessibility

Let

$$G = (V, E)$$

denote a network, let $S \subseteq V$ denote the current seed set, and let α denote the transmission probability under the Independent Cascade Model (ICM).

For each node $v \in V$, we first estimate its probability of receiving information:

$$p_v(S) = \Pr(v \text{ receives information} \mid S). \quad (1)$$

This quantity can be estimated using the PROBEST procedure employed by the Myopic algorithm, or using a computationally cheaper approximation such as a BFS- or PPR-based accessibility estimator.

I will suggest that you estimate it using the BFS- or PPR-based accessibility estimator. The first seed node should be selected using the highest pagerank pr highest core number. For the initial implementation, you can only use PageRank

We transform the information-access probability into a *disadvantage score*:

$$\boxed{D(v) = 1 - p_v(S)} \quad (2)$$

where $D(v) \in [0, 1]$. A high value of $D(v)$ indicates that node v currently has poor information access. For example,

$$p_v(S) = 0.1 \implies D(v) = 0.9,$$

whereas

$$p_v(S) = 0.8 \implies D(v) = 0.2.$$

Thus, $D(v)$ captures the first component of the seed-selection problem:

How underserved is node v under the current diffusion process?

1.2 Step 2: Estimating the Spreading Power of a Node

Accessibility alone is insufficient for seed selection because a highly disadvantaged node may have little ability to propagate information further. We therefore introduce a second quantity, $R(v)$, representing the *spreading power* of node v .

A correct way to define is the expected cascade size when v is selected as a seed:

$$R(v) = \mathbb{E}[|\text{Cascade}(v)|]. \quad (3)$$

However, conventional spreading power may not be desirable for fair information outreach as well as it is costly to compute, **so I propose a better estimator to estimate it.**

Fair Spreading Power To address this issue, we define a *fair spreading power* that gives greater importance to propagation toward poorly accessible nodes.

Let

$$P_{v \rightarrow u}$$

denote the probability that information originating from candidate node v reaches node u . We define

$$\boxed{R(v) = \sum_{\substack{u \in V \\ u \neq v}} D(u) P_{v \rightarrow u}} \quad (4)$$

or equivalently,

$$R(v) = \sum_{\substack{u \in V \\ u \neq v}} (1 - p_u(S)) P_{v \rightarrow u}. \quad (5)$$

Under this formulation, propagating information to a node with low current access contributes more to $R(v)$ than propagating information to an already well-served node.

Therefore, $R(v)$ answers the second question:

How effectively can node v propagate information toward currently underserved individuals?

Approximation of Fair Spreading Power Computing $P_{v \rightarrow u}$ exactly for every pair (v, u) can be computationally expensive. We therefore propose a simple distance-based approximation.

Under the Independent Cascade Model, transmission along a path of length d requires approximately d successful transmission events. Assuming a uniform edge transmission probability α , we use

$$P_{v \rightarrow u} \approx \alpha^{d(v,u)}, \quad (6)$$

where $d(v, u)$ denotes the shortest-path distance between v and u .

The fair spreading score can then be approximated as

$$R(v) = \sum_{\substack{u \in V \\ u \neq v}} D(u) \alpha^{d(v,u)} \quad (7)$$

This formulation has an intuitive interpretation:

- disadvantaged nodes receive larger weights through $D(u)$;
- nearby nodes receive larger weights because $\alpha^{d(v,u)}$ decreases with distance;
- distant nodes contribute less because reaching them requires more successful transmissions.

For improved scalability, propagation can be restricted to nodes within L hops:

$$R^{(L)}(v) = \sum_{\substack{u \neq v \\ d(v,u) \leq L}} D(u) \alpha^{d(v,u)} \quad (8)$$

where L is typically a small integer such as 2 or 3. **I will suggest that you do experiments for 2 or 3 only.** This avoids computing the contribution of distant nodes whose influence is already strongly discounted by $\alpha^{d(v,u)}$.

Normalization: don't do it for now I will suggest that in initial experiments, you can avoid the normalization step and just see what the range of these values is, and whether we can define lambda in such a way that we still get good results.

Since the accessibility-disadvantage score and spreading-power score may have different scales, we can normalize both quantities before combining them.

The normalized disadvantage score is

$$\tilde{D}(v) = \frac{D(v) - D_{\min}}{D_{\max} - D_{\min}}, \quad (9)$$

where

$$D_{\min} = \min_{u \notin S} D(u), \quad D_{\max} = \max_{u \notin S} D(u).$$

Similarly,

$$\tilde{R}_F(v) = \frac{R_F(v) - R_{F,\min}}{R_{F,\max} - R_{F,\min}}. \quad (10)$$

Consequently,

$$\tilde{D}(v), \tilde{R}_F(v) \in [0, 1].$$

1.3 Step 3: Final Score: Combined Propagation-Aware Fairness Score

We combine the two components using a weighted objective:

$$\boxed{FS(v) = \lambda \tilde{D}(v) + (1 - \lambda) \tilde{R}_F(v)} \quad (11)$$

where

$$0 \leq \lambda \leq 1.$$

The parameter λ controls the trade-off between prioritizing the candidate's own information disadvantage and its ability to improve information access for other disadvantaged nodes.

When

$$\lambda = 1,$$

the objective becomes

$$FS(v) = \tilde{D}(v),$$

which is equivalent in spirit to the Myopic strategy: preference is given entirely to nodes with poor current information access.

When

$$\lambda = 0,$$

the objective becomes

$$FS(v) = \tilde{R}_F(v),$$

and seed selection is determined entirely by the node's ability to propagate information toward disadvantaged individuals.

Intermediate values of λ balance these two objectives.

1.4 Step 4: Seed Selection

At each iteration, the candidate with the largest combined score is selected:

$$\boxed{v^* = \arg \max_{v \in V \setminus S} FS(v)} \quad (12)$$

The selected node is added to the seed set:

$$S \leftarrow S \cup \{v^*\}. \quad (13)$$

After adding the new seed, the information-access probabilities $p_v(S)$ are updated, which in turn updates the disadvantage scores $D(v)$ and fair spreading-power scores $R_F(v)$. The procedure is repeated until the desired seed budget k is reached.

2 Alternative approaches for final score

I propose 2 more approaches to compute the final score as defined in Step 3, and you can try all 3 of these and see which one is better.

2.1 Multiplicative Alternative

An alternative to the weighted additive objective is a multiplicative formulation:

$$\boxed{FS_{\text{mult}}(v) = \tilde{D}(v)^\lambda \tilde{R}(v)^{1-\lambda}} \quad (14)$$

A particularly simple version is

$$FS_{\text{mult}}(v) = \tilde{D}(v) \tilde{R}(v). \quad (15)$$

The multiplicative formulation favors candidates that perform well on *both* dimensions. For example, an extremely disadvantaged but structurally isolated node receives a low score because its spreading power is small. Similarly, a highly influential hub that already has very high information access receives a low score because its disadvantage is small.

In contrast, the additive formulation permits a sufficiently large value on one dimension to compensate for a smaller value on the other. We therefore consider the additive and multiplicative formulations as two variants of the proposed method.

2.2 Pareto-Based Candidate Filtering

Before computing the final combined score, an optional Pareto filtering step can be applied. Each candidate v is represented by the pair

$$\left(\tilde{D}(v), \tilde{R}(v) \right).$$

A candidate v is dominated by candidate u if

$$\tilde{D}(u) \geq \tilde{D}(v) \quad (16)$$

and

$$\tilde{R}_F(u) \geq \tilde{R}_F(v), \quad (17)$$

with at least one strict inequality. Dominated candidates can be removed because another candidate is simultaneously more disadvantaged and has greater fair spreading capability.

The combined scoring function can then be applied only to candidates on the Pareto frontier.

3 Proposed Algorithm

I roughly type the algorithm as it can help you in better understanding.

Algorithm 1 Propagation-Aware Fair Seed Selection

Require: Graph $G = (V, E)$, initial seed set S , seed budget k , transmission probability α , trade-off parameter λ , propagation depth L

Ensure: Seed set S

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1: add first seed node to  $S$  based on highest Pagerank
2: while  $|S| < k$  do
3:   Estimate information-access probability  $p_v(S)$  for every  $v \in V$ 
4:   for each  $v \in V \setminus S$  do
5:      $D(v) \leftarrow 1 - p_v(S)$ 
6:   end for
7:   for each  $v \in V \setminus S$  do
8:
```

$$R_F(v) \leftarrow \sum_{\substack{u \neq v \\ d(v,u) \leq L}} D(u) \alpha^{d(v,u)}$$

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9:   end for
10:  Normalize  $D(v)$  to obtain  $\tilde{D}(v)$  (we skip it for now)
11:  Normalize  $R_F(v)$  to obtain  $\tilde{R}_F(v)$  (we skip it for now)
12:  for each  $v \in V \setminus S$  do
13:
```

$$FS(v) \leftarrow \lambda \tilde{D}(v) + (1 - \lambda) \tilde{R}_F(v)$$

We try all 3 approaches for this

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14:  end for
15:
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$$v^* \leftarrow \arg \max_{v \in V \setminus S} FS(v)$$

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16:   $S \leftarrow S \cup \{v^*\}$ 
17: end while
18: return  $S$ 

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4 Interpretation

The proposed method separates seed selection into two complementary questions:

$$\underbrace{D(v)}_{\text{How underserved is node } v?} \quad \text{and} \quad \underbrace{R_F(v)}_{\text{How effectively can } v \text{ improve outreach to underserved nodes?}} \quad (18)$$

The final seed-selection criterion combines these quantities:

$$\boxed{\text{Fair Seed Score} = \lambda \underbrace{\text{Accessibility Disadvantage}}_{\text{individual need}} + (1 - \lambda) \underbrace{\text{Fair Spreading Power}}_{\text{downstream benefit}}.} \quad (19)$$

Thus, unlike Myopic, the proposed method does not automatically select the least-accessible node. Instead, it favors nodes that are themselves relatively underserved while also occupying network positions from which information can propagate effectively toward other underserved individuals.

First try for one setting and see how this method works, and if it is better or close to the best one, we can fine-tune it. Instead of doing experiments for all settings, first do them for minimal settings and share the results.